

PHOTOGRAPHY & MEDIA · PORTFOLIO WEBSITE

Hiccupviews

Technical & Design Justification

Project: Hiccupviews portfolio website (V2)

Course: Fontys — Media Creation

Document: Tech stack overview & design rationale

Date: June 2026

01 Introduction

Hiccupviews is a photography and media brand. This website is its digital storefront — a place to present work, communicate the services offered, and convert visitors into booking enquiries.

This document explains the **technology stack** behind the site and **justifies the reasoning** behind the technical and design decisions. The guiding principle throughout was simple: the photography is the product, so every choice — from the framework to the colour palette — should get out of its way and let the images carry the experience.

The site is a modern single-page application (SPA): a fast, app-like front end that loads once and then navigates instantly between pages, with the visual polish (animation, interactive hero, lightbox viewer) expected of a creative portfolio in 2026.

02 Technology Stack

The stack is deliberately lean and industry-standard. Each tool earns its place; nothing is included "just in case."

TECHNOLOGY	VERSION	ROLE
React	18.3	UI library — component-based interface
TypeScript	5.6	Typed JavaScript — safer, self-documenting code
Vite	5.4	Build tool & dev server
React Router	6.28	Client-side routing between pages
Framer Motion	12.38	Animation engine (scroll hero, transitions)
Tailwind CSS	3.4	Utility classes (used selectively)
PostCSS + Autoprefixer	8.4 / 10.4	CSS processing & vendor prefixing
lucide-react	0.460	Lightweight icon set
Custom CSS design system	—	Hand-built tokens & component styles

Why these choices

React is the most widely used front-end library, which makes it a safe, future-proof, and highly employable skill to build on. Its component model lets the interface be split into small, reusable,

testable pieces — the navigation bar, footer, photo lightbox and animated hero are each self-contained components reused across the site.

TypeScript adds a type layer on top of JavaScript. For a project with structured data (photo collections, hero slides, album metadata), types catch mistakes before the site is ever opened in a browser and act as living documentation. The whole project compiles type-clean as part of every production build.

Vite was chosen over older bundlers (such as Create React App / webpack) for speed. It offers near-instant hot-reload during development and produces a small, optimised, tree-shaken production bundle. A full production build completes in under two seconds, which keeps the iteration loop tight.

React Router provides client-side navigation between the Home, Portfolio, Album, Services, About and Contact pages. Visitors move between pages instantly with no full page reload, giving the site an app-like feel while keeping clean, shareable URLs.

Framer Motion powers the signature scroll-driven hero animation and the page micro-interactions. It is a declarative, physics-based animation library — animations are described as states rather than hand-written timing code, which keeps complex motion (the scatter → line → circle → arc morph) maintainable.

Tailwind CSS is used *selectively* for utility styling, with its global reset deliberately disabled so it never clobbers the hand-built design system. This is a pragmatic hybrid: Tailwind for quick layout utilities, bespoke CSS for the brand's distinct visual identity.

A note on restraint. A photography site does not need a heavy framework like Next.js, a CMS, or a component library such as Material UI. Those would add weight, opinionated styling, and maintenance overhead the project does not need. The chosen stack is light enough to host as static files on any CDN, yet powerful enough for rich interaction.

03 Architecture & Structure

The codebase is organised by responsibility, which keeps it easy to navigate and extend:

- **Pages** — one component per route (Home, Portfolio, Album, Services, About, Contact).
- **Components** — shared building blocks: navigation, footer, the scroll hero, and the fullscreen lightbox photo viewer.
- **Data** — typed content files (photo collections, album metadata, hero captions) kept separate from presentation, so updating content never means touching layout code.
- **Hooks** — small reusable behaviours (e.g. scroll-position and scroll-reveal logic).
- **Styles** — a central design-token stylesheet plus one stylesheet per page/feature.

Styling is driven by **CSS custom properties** (variables) for colour, typography, spacing and motion. Because every colour and font flows from a single token block, a site-wide change — such as the recent switch from a dark theme to a clean white theme — is a small, low-risk edit rather than a hunt across dozens of files.

04 Design Reasoning

A white, minimal canvas

The site uses a white background with near-black text and a single warm taupe accent. This is intentional: a neutral, low-noise canvas maximises contrast against the photography and lets each image read clearly without competing with the interface. The minimalism is a feature, not an absence of design — it directs all attention to the work.

An interactive, scroll-driven hero

The home page opens with an animated hero: the photographs assemble on load and then morph as the visitor scrolls, while the accompanying captions cycle through the brand's services (Events, Portraits, Corporate, Brand, Festivals). This does three jobs at once — it makes a memorable first impression, it showcases the breadth of work immediately, and it communicates what the brand offers without a wall of text.

Frictionless viewing

Any photograph can be clicked to open in a fullscreen lightbox with keyboard arrows, swipe support and a counter. Visitors can study the actual work — the core purpose of the site — with zero friction.

Accessibility & inclusivity

Rich motion is balanced with care for the visitor:

- A **reduced-motion fallback** respects the operating-system setting and skips the scroll-hijacking animation for users who are sensitive to motion.

- A visible **skip control** lets anyone bypass the intro.
- Semantic headings, descriptive **alt text** on every image, and full keyboard support in the lightbox keep the site usable for assistive technology.

Performance

Photography sites live or die on load speed. The build serves an optimised, compressed bundle, and every photo exists in two sizes — a small thumbnail for grids and animation, and a full-resolution version loaded only when a visitor opens it. Images are lazy-loaded, so the browser never downloads more than it needs.

Responsive by design

Layouts, the hero geometry, and typography all adapt across phone, tablet and desktop using fluid sizing and breakpoints, so the experience holds up wherever the brand is discovered — which, for a creative business, is most often on a phone.

05 Why This Website Exists

Beyond the technology, the site serves clear goals for the Hiccupviews brand:

- **Credibility.** A polished, self-owned website signals professionalism in a way that a social-media feed alone cannot.
- **Showcase.** It presents the portfolio on the brand's own terms — full quality, no platform compression, no competing content.
- **Conversion.** Clear services and a direct booking/contact path turn interest into enquiries.
- **Ownership.** Unlike a third-party platform, the brand fully controls the presentation, the data and the experience.

Every technical and design decision documented above ladders up to these goals: a fast, beautiful, accessible site whose only job is to make the photography look as good as it deserves to.

o6 Conclusion

The Hiccupviews website pairs a lean, modern, industry-standard stack — React, TypeScript and Vite — with carefully chosen tools for routing, animation and styling. The result is a maintainable codebase and a fast, polished experience. The design language is deliberately restrained so that the photography remains the hero, while interactive touches and a strong attention to accessibility and performance give the site a distinctive, professional edge.

In short: **the right tools, used with restraint, in service of the work.**